

Helix VPN Design System

Helix VPN — Design System & UI/UX Documentation

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Overview

Enterprise-grade, cross-platform design system for Helix VPN. 8 platforms, 3 UI frameworks, unified design language, light+dark themes, customizable palettes, WCAG 2.1 AA accessibility, and bleeding-edge quality.

Quick Navigation

Document	Description
OpenDesign DESIGN.md	9-section design system spec (OpenDesign format)
OpenDesign tokens.css	Compiled CSS custom properties (light+dark)
OpenDesign manifest.json	Design system manifest
OpenDesign components.html	Interactive component reference
Component Library	Full component specs (all platforms)
Screen Wireframes	Every screen across all platforms
Interaction Patterns	Animation specs, gestures, accessibility

Design System Summary

1. Brand Identity

- **Name:** Helix VPN
- **Primary color:** Teal (#00897B) — trust, security, technological sophistication
- **Logo:** Shield/globe motif
- **Tagline:** “Your privacy starts here”
- **Voice:** Confident, calm, competent — security is serious, not playful

2. Platform Coverage

#	Platform	UI Framework	OS Versions	Bundle Target	Theme
1	macOS (Intel + Apple Silicon)	Tauri v2 + React	macOS 12+	.dmg Universal	System (light/dark)
2	Windows (x64 + ARM64)	Tauri v2 + React	Win 10/11	.msi / .exe	System (light/dark)
3	Linux (x64 + ARM64)	Tauri v2 + React	Ubuntu 22.04+	.AppImage / .deb / .rpm	System (light/dark)
4	Android (ARM64 + x86_64)	Flutter + Dart	API 26+ (8.0+)	.apk / .aab	Material You + Dynamic System (light/dark)
5	iOS (ARM64)	Flutter + Dart	iOS 15+	.ipa	System (light/dark)
6	HarmonyOS (ARM64)	Flutter + ohos	API 12+	.hap	System (light/dark)
7	Aurora OS (ARM64)	Qt6 / QML	Aurora 4.x+	.rpm	Silica Ambiance (dark-first)
8	Web Extension	MV3 + React	Chrome 90+, FF 88+, Edge 90+, Safari 15+	.zip	Browser

3. Technology Stack

Layer	Desktop	Mobile	Aurora	Web
UI Framework	Tauri v2	Flutter 3.29+	Qt6 / QML	React + MV3
Rendering	WKWebView/ WebView2/ WebKitGTK	Impeller	Silica (Qt)	Browser engine
State Management	Zustand / Valtio	flutter_bloc / Riverpod	C++ QObject	Zustand

Layer	Desktop	Mobile	Aurora	Web
Rust Bridge	Tauri Commands (IPC)	flutter_rust_bridge + UniFFI	Direct C FFI	wasm-bindgen
Icons	Lucide React	Material Symbols / SF Symbols	Silica icons	Lucide
Design Tokens	CSS custom properties	ThemeData / CupertinoTheme	QML properties	CSS vars

4. Design Token Summary

Domain	Count	Key Tokens
Colors (primary)	10	Primary-50 through Primary-900
Colors (semantic)	12	connected, connecting, disconnected, error, warning, info
Colors (surfaces)	8	bg-primary, bg-secondary, bg-tertiary, bg-elevated (× light/dark)
Typography	18	display, headline, title-lg/md/sm, body-lg/md/sm, caption, overline
Typography (mono)	4	mono-lg, mono-md, mono-sm, mono-caption
Spacing	12	2px through 80px (4px base grid)
Border radius	7	none through full (9999px)
Elevation	5	0–4 (flat → glow)
Motion	4	fast, base, slow, page (easing variants)

Theme System

5. Light + Dark Theme

Both themes are **authored** — every color token is explicitly set. Dark mode is NOT an automatic inversion.

Surface	Light	Dark
App background	#F0F4F8	#0A1929
Cards/panels	FFFFFF	#132F4C
Elevated (modal)	#E8EDF2	#1E4976
Modals/dialogs	FFFFFF	#243B55
Primary text	#0A1929	#F0F4F8
Secondary text	#4A6375	#8BA3B8

6. Custom Palette Architecture

The token system supports any brand seed color via CSS custom property override:

```
/* Built-in presets */
[data-theme="ocean-blue"] { --hx-primary-500: #2563EB; }
[data-theme="midnight"]   { --hx-primary-500: #7C3AED; }
[data-theme="forest"]     { --hx-primary-500: #059669; }
[data-theme="ruby"]       { --hx-primary-500: #E11D48; }
[data-theme="amethyst"]   { --hx-primary-500: #A855F7; }

/* User custom */
:root[data-theme="custom"] { --hx-primary-500: <user-color>; }
```

Theme/appearance setting: System (default) · Light · Dark · Custom (with palette picker)

Implementation Checklist

Phase 1: Foundation



Token architecture (colors, typography, spacing, elevation, motion, radii)



Light + dark theme definitions



Custom palette system (5 presets + user custom)



Font family selection per platform



Responsive breakpoints

Phase 2: Core Components



Button variants (primary, secondary, ghost, danger, icon)



Connection toggle with full state machine



Server list item



Protocol badge



Latency indicator



Input fields



Modal/dialog



Toast notification



Card component



Toggle/switch (adaptive per platform)

Phase 3: Specialized Components



Speed graph / sparkline



Data usage indicator



Bottom sheet (mobile)



Tab bar (adaptive)



Pull-to-refresh



Swipe actions (mobile)



System tray menu (desktop)



Quick Settings tile (Android)



Home screen widget (Android/iOS)



Aurora Silica components



Web extension popup components

Phase 4: Screen Layouts



Desktop main window (connected/disconnected)



- ☐ Desktop server selection
- ☐ Desktop settings (tabbed)
- ☐ Desktop connection details/stats
- ☐ Mobile connection home screen
- ☐ Mobile server selection
- ☐ Mobile settings
- ☐ Mobile split tunneling config
- ☐ Mobile connection stats
- ☐ Mobile support/help
- ☐ Aurora main view
- ☐ Aurora cover page
- ☐ Aurora pulley menu
- ☐ Web extension popup (connected/disconnected)
- ☐ Web extension options page
- ☐ Admin dashboard
- ☐ Onboarding flow

Phase 5: Interaction & Animation

- ☐ Connection flow animation (3-phase, 2200ms)
- ☐ Disconnect flow (800ms)
- ☐ Page transitions (per platform)
- ☐ Micro-interactions (16 defined)

- ☒ Reduced motion support
- ☒ Keyboard navigation (desktop)
- ☒ Gesture navigation (mobile)
- ☒ Haptic feedback patterns
- ☒ Loading states (skeleton, spinner, progress)
- ☒ Screen reader announcements

Phase 6: OpenDesign Deliverables

- ☒ DESIGN.md (9-section OpenDesign spec)
- ☒ tokens.css (canonical CSS custom properties)
- ☒ manifest.json (design system metadata)
- ☒ components.html (interactive reference)
- ☒ Component library documentation
- ☒ Screen wireframes documentation
- ☒ Interaction patterns specification
- ☒ Full design documentation index (this document)

Phase 7: Export & Validation

- ☐ Export DESIGN.md → PDF
- ☐ Export components.html → PNG screenshots
- ☐ Export all documentation to PDF/HTML
- ☐ Visual regression golden screenshots
- ☐ WCAG 2.1 AA contrast validation
- ☐

- ☐ Cross-platform consistency review
 - ☐ Figma design file generation
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Design Rules (Non-Negotiable)

1. **No overlapping elements.** Layout must never produce overlapping interactive elements or superimposed labels.
 2. **No color-only indicators.** Every status must pair color + icon + text.
 3. **No platform ignorance.** A Material-style switch on iOS is a defect — use CupertinoSwitch.
 4. **No decorative-only effects.** Every gradient, shadow, and animation serves a functional purpose.
 5. **No skeleton-only loading.** Show real content within 2s or error state.
 6. **No dark-mode inversion.** Dark theme is explicitly authored (every [data-theme="dark"] token).
 7. **No hardcoded thresholds.** Every latency/spacing/timing constant reads from tokens.
 8. **No accessibility shortcuts.** WCAG 2.1 AA minimum enforced on every screen.
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File Structure

docs/design/	
├─ README.md	← This file (master index)
├─ opendesign/	
│ └─ helix/	
│ └─ manifest.json	← OpenDesign design system manifest
│ └─ DESIGN.md	← 9-section OpenDesign spec
│ └─ tokens.css	← Compiled CSS custom properties
│ └─ components.html	← Interactive component reference
│ └─ assets/	← Design assets (logos, icons, graphics)
│ └─ preview/	← Preview renders
│ └─ source/	← Source files
├─ tokens/	
│ └─ color.json	← Color token definitions
│ └─ typography.json	← Type scale token definitions
│ └─ spacing.json	← Spacing scale definitions
│ └─ ...	← Per-category token defs
├─ components/	
│ └─ README.md	← Component library master
│ └─ desktop/	← Desktop-specific components

		mobile/	← Mobile-specific components	
		aurora/	← Aurora-specific components	
		web/	← Web-specific components	
		screens/		
			README.md	← All screen wireframes
			desktop/	← Desktop screen specs
			mobile/	← Mobile screen specs
			aurora/	← Aurora screen specs
			web/	← Web extension screen specs
		interaction/		
			README.md	← Animations, gestures, accessibility
		icons/		
			app-icon/	← App icon variants (all sizes)
			tray-icon/	← System tray icon per platform
			extension-icon/	← Browser extension icon
		exports/		
			HelixVPN-Design-System.pdf	← Full design system export
			HelixVPN-Component-Library.pdf	← Component library export
			HelixVPN-Screen-Wireframes.pdf	← Screen wireframes export
			HelixVPN-Interaction-Specs.pdf	← Interaction patterns export